1) Verificar si $y = 1$ es solución particular de la ecuación $\frac{dy}{dx} + (2x-1)y - xy^2 = x - 1$. En caso afirmativo, calcular la solución general. $Rta.: y = \frac{e^x(2-x)+c}{e^x(1-x)+c}$

$$y_0 = 1$$

$$y = 1 + z \rightarrow z = y - 1$$

$$\frac{dy}{dx} = \frac{dz}{dx}$$

$$\frac{dz}{dx} + (2x-1)(1+z) - x(1+z)^2 = x - 1$$

$$\frac{dz}{dx} + 2x + 2xz - 1 - z - x(1 + 2z + z^2) = x - 1$$

$$\frac{dz}{dx} + \cancel{2x} + \cancel{2xz} - \cancel{1} - z - \cancel{x} - \cancel{2xz} - z^2 x = \cancel{x} - \cancel{1}$$

$$\frac{dz}{dx} - z = z^2 x$$

$$z^{-2}\frac{dz}{dx} - z^{-1} = x$$

$$z^{-1} = t$$

$$\frac{dt}{dx} = -z^{-2}\frac{dz}{dx}$$

$$-\frac{dt}{dx} - t = x$$

$$\frac{dt}{dx} + t = -x$$

$$t = ul$$

$$\frac{dt}{dx} = u\frac{dl}{dx} + l\frac{du}{dx}$$

$$u\frac{dl}{dx} + l\frac{du}{dx} + ul = -x$$

$$u\left(\frac{dl}{dx} + l\right) + l\frac{du}{dx} = -x$$

$$\frac{dl}{dx} + l = 0$$

$$\frac{dl}{dx} = -l$$

$$\frac{dl}{l} = -dx$$

$$\ln l = -x$$

$$l = e^{-x}$$

$$e^{-x}\frac{du}{dx} = -x$$

$$du = -x e^{x} dx$$

$$u = \frac{xe^{x}}{2} + c$$

$$t = \left[\frac{xe^{x}}{2} + c_1\right] e^{-x}$$

$$t = \frac{x}{2} + C$$

$$z^{-1} = \frac{x}{2} + C$$

$$\frac{1}{z} = \frac{x}{2} + C$$

$$\frac{1}{y-1} = \frac{x}{2} + C$$

$$\frac{1}{y-1} = \frac{x+2C}{2}$$

$$y-1 = \frac{2}{x+2C}$$

$$y = \frac{2}{x+2C} + 1$$

$$y = \frac{2+x+2C}{x+2C}$$

$$\int v\,du = uv - \int u\,dv$$

$$u = x \qquad du = dx$$

$$dv = e^{x}dx \qquad v = e^{x}$$

$$\int xe^{x}dx = xe^{x} - \int xe^{x}dx$$

$$2\int xe^{x}dx = xe^{x}$$